

The Recognition Lag

Who Gets Recognized, and When?

A Data Analysis of Acquisition Lag in the MoMA Collection

Workintech Data Science Bootcamp — Capstone Project

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160,699 artworks · 15,932 artists · BigQuery + dbt + Python · Wikidata enrichment

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Abstract

This study examines the acquisition behavior of the Museum of Modern Art (MoMA) using its collection data. The core metric is “acquisition lag”: the difference between the year a work was produced and the year it entered the MoMA collection. The MoMA GitHub dataset of 160,699 artworks and 15,932 artists was enriched with movement and geography data obtained via Wikidata SPARQL queries, and processed through a layered dbt architecture on BigQuery. Four hypotheses were tested: (H1) gender, (H2) geographic origin (Western / non-Western), (H3) art movement, and (H4) the temporal consistency of MoMA's acquisition behavior. Descriptive findings were validated through confounding-variable control (period-based comparison) and inferential statistics (Mann-Whitney U, Pearson correlation). The results show that the gender-based acquisition-lag gap has closed statistically ($p=0.418$, 1990–2010), while the gap based on geographic origin has remained significant across all periods ($p<0.001$). The movement analysis found that MoMA focused on its own contemporary movements early on, and in later periods collected in a way that compensated for movements it had previously neglected (the direction of the correlation reversed from $r=-0.64$ in 1940–1970 to $r=+0.73$ in 2000–2030). The institution's acquisition behavior shows strong temporal consistency ($r=0.855$, $p=0.0016$). Finally, a regression model predicting an artist's total output from their productivity in the first eight years of their career ($R^2=0.705$) was built, and used to estimate the number of works potentially missing from the MoMA collection for 187 artists who died young.

1. Introduction

A museum collection is not a neutral record of art history; it is the accumulation of a series of institutional choices. A museum cannot acquire every work that exists — every work it does acquire is a decision about who, when, and from where it collected. The starting point of this study is the question: “Who do museums recognize, and when?”

To answer this question, the metric of “acquisition lag” — the difference between the year a work was produced and the year it entered the MoMA collection — is placed at the center. This metric is a verifiable target variable that exists directly in the data, and is examined along three dimensions:

- The institution's response time — the overall distribution of acquisition lag and how it has changed over time
- Equity and visibility — whether the lag differs by gender, geography, and art movement
- Institutional behavior pattern — whether MoMA follows the market or acts as a pioneer, and how consistent this behavior has been over time

The core methodological discipline adopted in the project is to run every finding through a three-stage filter: (1) descriptive/raw observation, (2) confounding-variable control (in particular, period-based comparisons that isolate the effect of the production-year gap), and (3) inferential statistical validation. As will be seen repeatedly in later sections of the report, this approach revealed that some findings that seem correct at first

glance (e.g. the raw version of H1) can be misleading, while others (H2) are genuinely persistent beyond the first impression.

2. Data and Methods

2.1 Data Sources

The study's core data source is the collection dataset MoMA publishes directly as CSV under a CC0 license on GitHub (github.com/MuseumofModernArt/collection). This dataset consists of two main tables:

- **Artworks.csv** — 160,699 artwork records: title, production date (Date), acquisition date (DateAcquired), medium, classification, department, dimensions, and credit line (gift, bequest, purchase).
- **Artists.csv** — 15,932 artist records: display name, nationality, gender, birth year (BeginDate), death year (EndDate), Wikidata QID, and ULAN identifier.

This core data was enriched via SPARQL queries against artists' Wikidata identifiers (QID). Three properties were pulled from Wikidata: P135 (movement — e.g. Cubism, Surrealism, Bauhaus), P19 (place of birth — distinct from nationality, the field actually used for the geography question), and P27 (citizenship). For the 3,224 artists with a QID, an enrichment table of 4,877 rows in total (`moma_raw.wikidata_enrichment`) was built. Since the artist–movement relationship is one-to-many (an artist can be associated with more than one movement), this enrichment required a bridge table.

An important scope limitation: not every artist has a Wikidata record. The movement (H3) analysis was therefore carried out not on the full collection but on a subset of ~69,755 artworks (43.4%) whose movement is known. This subset is not random — artists with a Wikidata record tend to be relatively better-known, canonical artists; this means the H3 findings may carry a selection bias, which is discussed further in the relevant section.

2.2 Data Cleaning

The field requiring the most work in the raw data was the Date column (production year). Patterns found together in this single column included clean years (“1957”), estimated years (“c. 1960”), year ranges (“1962-64”), double-dated records (“1935 (printed 1970)”), abbreviations (“n.d.”), decade notations (“1970s”), and missing/unknown values (“Unknown”, empty cells). The method followed had four steps: (1) classify the patterns with regex and count how many records fell into each pattern, (2) define an explicit rule for each pattern (e.g. a “c.” prefix is treated as an estimate, and range records take the start year), (3) document the decisions in a rules table, and (4) never silently drop a row — report the loss instead. Three rounds of regex classification in BigQuery reduced the number of records in the “other” category from 30,773 to 12,382, after which the `extract_year_info` function written in Python was able to extract a valid year for 97.25% of the 160,699 rows.

Other cleaning steps included the following: the Gender field was reduced from 428 free-text categories to 5 standard categories (male, female, non-binary/other, unknown, ambiguous); MoMA's own ambiguity notes

such as “male (trans? ftm?)” were preserved as a separate category. Since the Nationality field contained 145 distinct demonym values, including states that no longer exist (“Yugoslav,” “Czechoslovakian,” “Soviet”), a manual/AI-assisted country–region mapping dictionary was built to translate these into geographic regions. Where a work had more than one artist, the artist list was unpacked into rows with SPLIT+UNNEST, producing an artwork–artist relationship table of 176,340 rows (including 8,109 multi-artist works).

2.3 Data Architecture

The project was run on a layered dbt architecture on BigQuery following the single-source-of-truth principle:

- staging — the layer where the cleaning logic on the source tables (date parsing, category reduction) is applied (stg_artworks, stg_artists, stg_wikidata_enrichment, stg_uyruk_cografya_sozlugu).
- intermediate — the bridge tables where artwork–artist–movement relationships are built (int_artwork_artist_link, int_artwork_artist_detail, int_artwork_artist_movement).
- marts — the final analysis-ready tables: mart_acquisition_lag (the core metric), mart_gender_analysis, mart_geography_analysis, mart_movement_analysis.

All dbt models were committed to GitHub under version control. Analysis results (summary tables, period-based comparisons, model outputs) were additionally consolidated into a separate BigQuery dataset called moma_dashboard, making them directly usable at the dashboard stage without needing to be recomputed from scratch.

2.4 Tools Used

Google BigQuery (data warehouse and single source of truth), dbt (transformation layers and testing), Python / Google Colab (EDA, statistical testing, and machine learning with pandas, numpy, matplotlib, seaborn, scipy, scikit-learn), GitHub (dbt project version control). Automation (e.g. Fivetran, scheduled data pipelines) was deliberately not used: since the project relies on fixed, one-time sources (static CSV files, a one-time SPARQL pull), there was no need for a dynamic data pipeline.

2.5 Hypotheses

Code	Topic	Question
H1	Gender	Does acquisition lag differ between works by female and male artists, and if so, is the gap real or a chronological illusion?
H2	Geography	Is the gap between Western and non-Western artists persistent or temporary?
H3	Art Movement	Which movements did MoMA recognize early, and which late — and has this pattern changed over time?
H4	Temporal Consistency	Is MoMA's acquisition behavior predictable, or does it change randomly from period to period?

3. Findings

3.1 Overall Acquisition Lag

Across the 151,146 artworks with computable date information, the median acquisition lag is 21 years and the mean is 29.3 years (std=29.1; min=0; max=196). The gap between the mean and the median shows the distribution is right-skewed: a small number of very-late-acquired works pull the mean upward, while the median better represents typical behavior.

Examined by decade, the 1960s stand out as a peak both in median lag (22 years) and in artwork count (43,107 — many times higher than any other decade). The mechanism behind this peak was explained by an anomaly noticed during pairplot exploration (see Section 4.1): in 1964 and 1968, 12,987 and 19,010 artworks respectively were acquired in a single year, and these two years appear to consist almost entirely (97–99%) of works by Western male artists in the form of massive bulk donations. The median lag stayed relatively stable in the 7–10-year range between 1970 and 1990 (a period when “fresh,” not-yet-canonized contemporary art was collected quickly), and rose continuously after 2000 (1990: 10 years → 2000: 27 → 2010: 38 → 2020: 47).

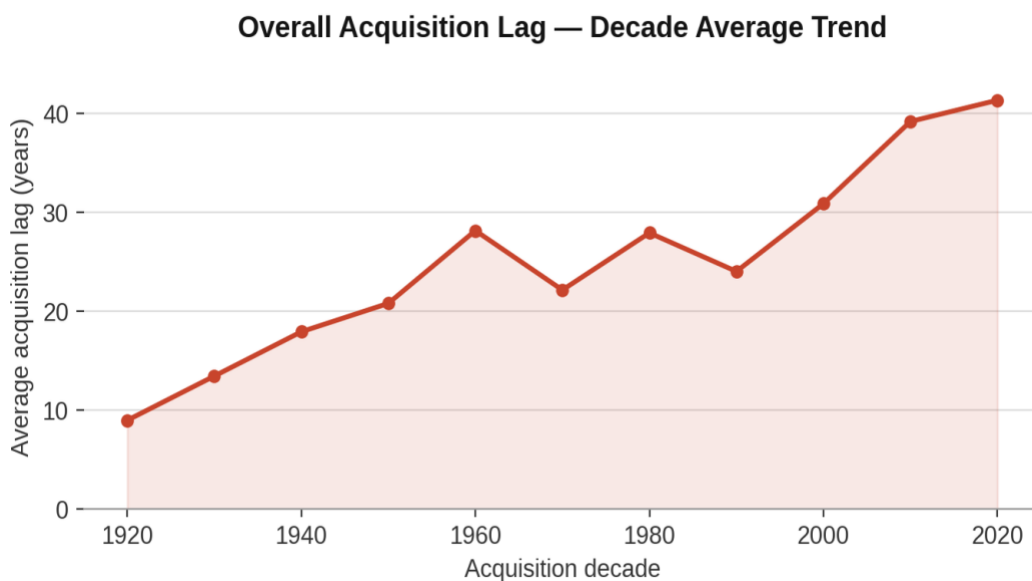


Figure 1. Decade-by-decade average acquisition lag trend.

3.2 H1 — Gender

Overall gender distribution of the collection: male 83.0%, female 14.2%, unknown 2.8%.

The raw (uncontrolled) comparison showed that works by female artists were accepted faster than those by male artists. However, this is a classic confounding-variable effect stemming from the fact that the average production year of works by female artists is noticeably more recent than that of works by male artists (a gap of roughly 23 years): because less time had passed, the lag appears lower too. A period-based comparison that holds production year constant reveals the real pattern:

Period	Female median (yr)	Male median (yr)	Mann-Whitney p	Result
1950-1970	39	11	< 0.001	Significant

Period	Female median (yr)	Male median (yr)	Mann-Whitney p	Result
1970-1990	27	18	< 0.001	Significant
1990-2010	5	5	0.418	Not significant

Table 1. H1 period-based median acquisition lag and Mann-Whitney U test results.

This result completely reverses H1's raw finding: works by female artists were, historically, accepted significantly more slowly ($p < 0.001$) between 1950 and 1990; this gap became statistically non-significant and closed completely by 1990–2010.

As a robustness check, the analysis was repeated with the two mega-donation years, 1964 and 1968, removed from the 1950–1970 window; the female–male median gap narrowed from 28 years to 25 years but was largely preserved. This confirms that the H1 finding does not depend on a single unusual event, and that it is a persistent pattern present in MoMA's general acquisition behavior for the period.

Artwork-count projection: even though the lag behavior has improved, the ratio of female to male artwork counts in the collection is still unbalanced (as of 2020: 21,828 works by women / 123,378 works by men). A year-by-year crossover simulation using 1950–2010 period-based growth rates (with the 1960 outlier removed) shows that, if the current trend continues, the cumulative artwork counts would equalize around the year 2097.

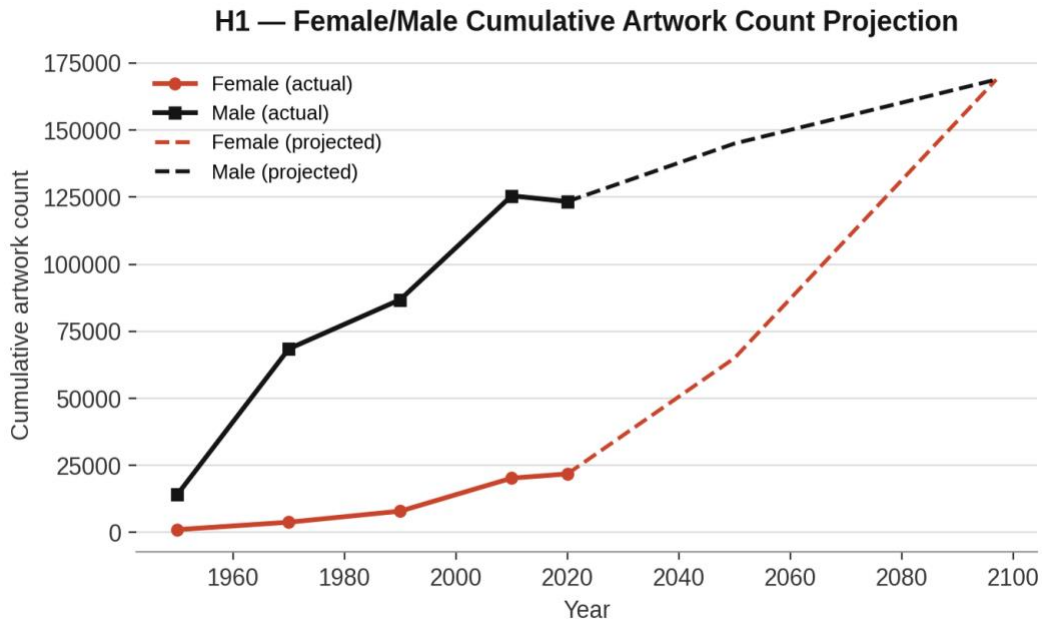


Figure 2. H1 — female/male cumulative artwork count projection.

3.3 H2 — Geography (Western / Non-Western)

Artists from European and North American countries were classified as “Western,” all others as “non-Western.” Overall distribution: Western 85.3%, non-Western 13.9%, unknown 0.8%.

Unlike H1, the production-year gap here is small (~9 years) — pointing to a lag gap that cannot be explained by chronology. A period-controlled comparison confirms this:

Period	Western median (yr)	Non-Western median (yr)	Mann-Whitney p	Result
1950-1970	11	39	< 0.001	Significant
1970-1990	15	33	< 0.001	Significant
1990-2010	4	9	< 0.001	Significant

Table 2. H2 period-based median acquisition lag and Mann-Whitney U test results.

Unlike H1, the gap remained extremely significant statistically across all three periods. The gap narrowed (28 years in 1950–70, 5 years in 1990–2010) but never lost statistical significance in any period. This shows that geography-based inequality is a far more persistent, systematic pattern in MoMA's acquisition practice compared to gender.

On the artwork-count side, the number of Western works declined over time (34,034 → 24,678 → 17,045, by period) while the number of non-Western works stayed relatively stable (4,887 → 4,350 → 4,275), meaning the proportional share is growing in favor of non-Western works. A projection using the same method as H1 shows that with 18,969 non-Western works and 127,205 Western works as of 2020, the cumulative counts could equalize around 2089 at the current trend.

The fact that these two projections (2097 and 2089) come out numerically close should not be misleading: statistical tests show the two inequalities are qualitatively different. In H1 the lag gap closed completely and statistically (a structural behavioral improvement), while in H2 the lag gap never closed in any period (a persistent pattern); the similarity of the artwork-count projections is purely coincidental, resulting from the two growth rates happening to be close.

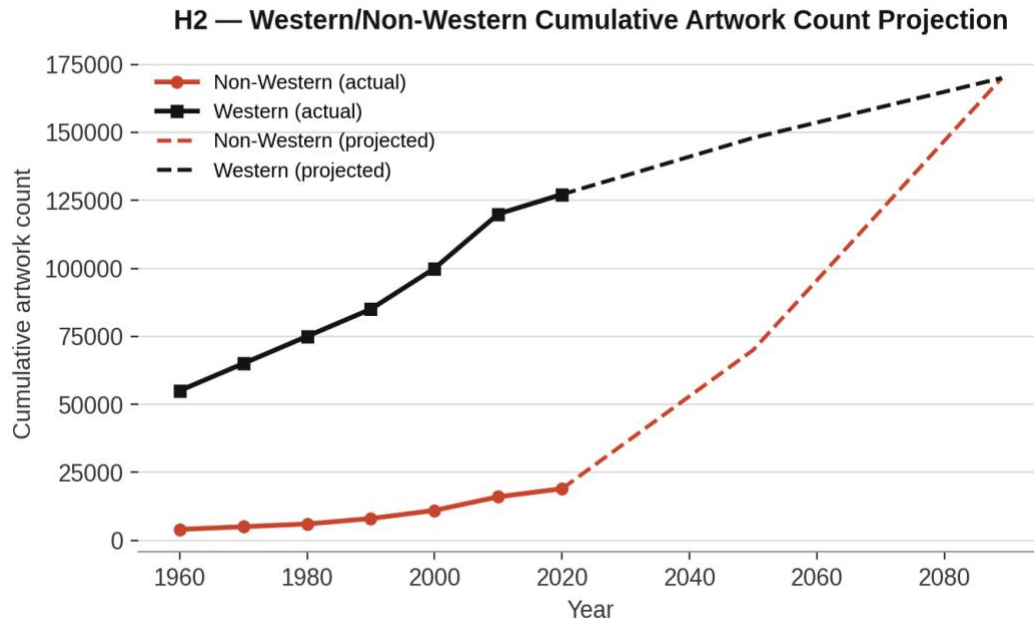


Figure 3. H2 — Western/non-Western cumulative artwork count projection.

3.4 H3 — Art Movement

In the subset of ~69,755 artworks with a known movement, the collection's most common movements are, as expected, 20th-century modern art movements: modernism (16,854 works, 10.5% of the collection), Neues Bauen (13,136, 8.2%), surrealism (8,119, 5.1%), abstract expressionism (5,920, 3.7%), Expressionism (5,715, 3.6%).

A trend line was fit between each movement's average production year and its average acquisition lag, and the deviation from this line ("recognized earlier/later than expected") was calculated. To improve the reliability of the trend line, five historical reference movements (Rococo, Romanticism, literary realism, Art Nouveau, Orientalism) were added to the first 10 movements.

Movement	Deviation (yr)	Interpretation
Expressionism	-19.3	Recognized far earlier than expected
Neues Bauen	-15.8	Recognized earlier than expected
modernism	-10.7	Recognized earlier than expected
Rococo / Romanticism / Art Nouveau	≈ 0	Close to the trend line — lag explained by chronology
abstract art	+11.6	Recognized later than expected
feminist art	+11.8	Recognized later than expected
Minimalism	+18.5	The most-belatedly-recognized movement

Table 3. Deviation of selected movements from expected acquisition lag.

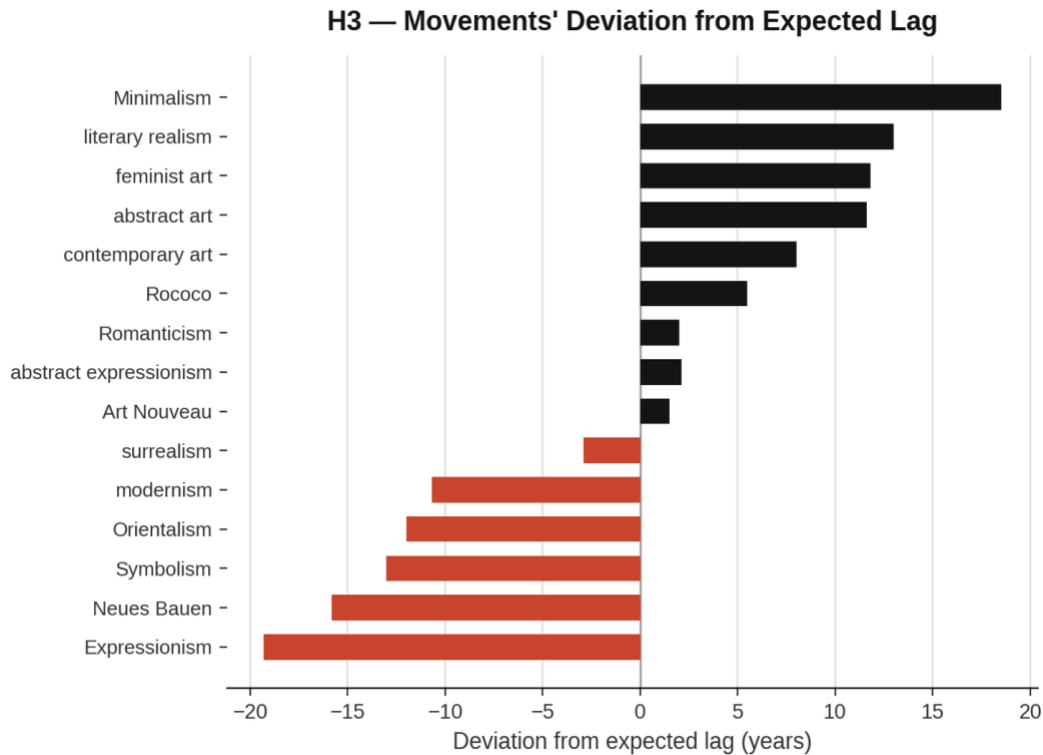


Figure 4. Movements' deviation from expected lag (all 15 movements).

It is notable that historical movements like Rococo and Romanticism, despite having a high raw lag (spanning over a century), come out close to the trend line (a small deviation): this confirms that their high lag is fully explained by chronology, not by neglect.

The overall correlation between deviation and collection weight (artwork count) was not statistically significant ($r=-0.426$, $p=0.113$, $n=15$ — a small sample). However, breaking the data down by period showed that MoMA's “recognize-early / collect-heavily” behavior reversed direction over time, and that this reversal was statistically significant:

Period (acquisition)	Correlation (r)	p-value	Result
1940-1970	-0.64	0.015	Significant — early-recognized movements dominant
1970-2000	+0.14	0.633	Not significant — transition period
2000-2030	+0.73	0.006	Significant — pattern reversed

Table 4. Period-based change in the deviation $\times$ collection-weight correlation.

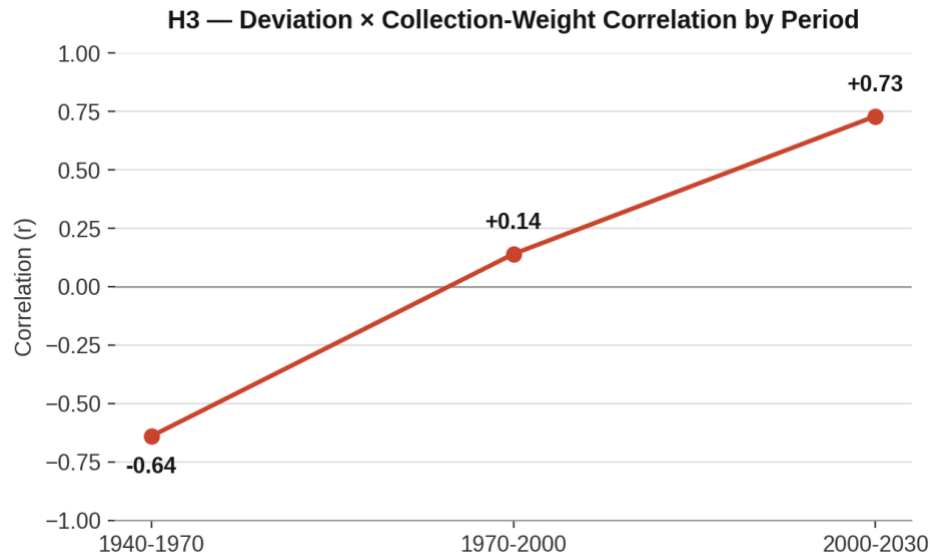


Figure 5. Period-based change in the deviation × collection-weight correlation.

Interpretation: during its founding period (1940–1970), MoMA focused heavily on movements that were its own contemporaries (modernism, Neues Bauen, Expressionism); after 2000, this pattern reversed, and movements that had previously been relatively neglected (Minimalism, feminist art, abstract art, contemporary art) began gaining weight in the collection. The acquisition trend for these four movements between 1990 and 2010 is consistently positive (growth of +11 to +58), making them the most reliable candidates for MoMA's “catch-up” behavior.

3.5 H4 — Temporal Consistency

The relationship between the previous period's lag and the current period's lag (lag-1 autocorrelation) was tested on the decade-by-decade average lag series: $r=0.855$, $p=0.0016$ ($n=10$). This is a strong and highly statistically significant relationship.

The finding shows that MoMA's acquisition behavior is not random/unpredictable, but rather carries strong temporal consistency: the level of lag in one period is largely preserved in the next. A 2100 projection using the autocorrelation trend shows the average lag approaching an equilibrium of around 45 years (2090: 44.9 years; 2100: 45.2 years — the rate of increase is clearly slowing).

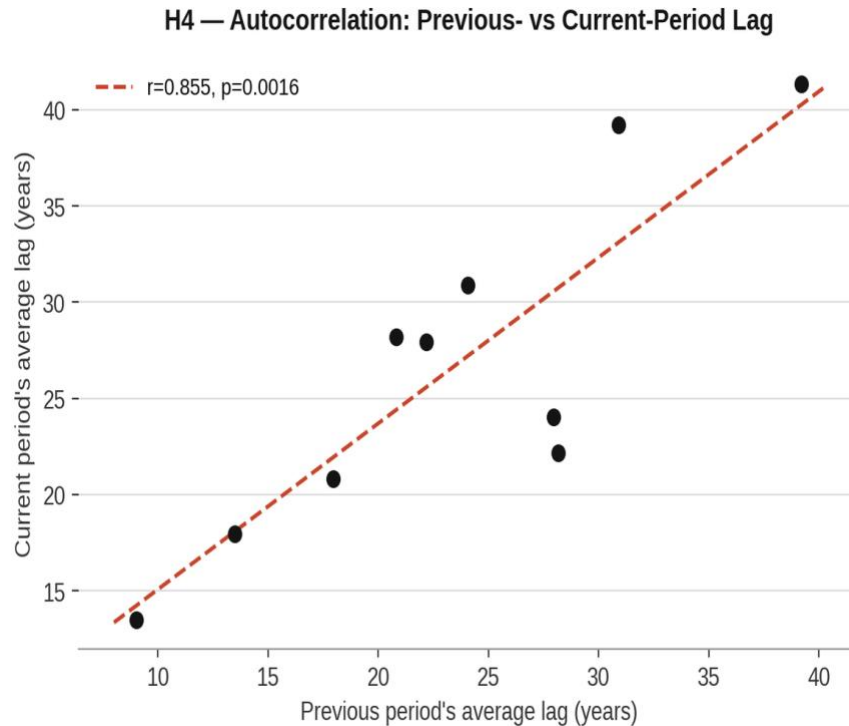


Figure 6. H4 — autocorrelation between the previous period's and the current period's lag.

3.6 Additional Model — Early-Death Artist Projection

This additional analysis asks: can an artist's total lifetime output be predicted from their productivity in the early years of their career? If the model is accurate, it can be applied to artists who died young to produce an estimate for the question “how many works would potentially be in the collection today if they had lived a normal lifespan?”

Using the BeginDate/EndDate fields in Artists.csv, 5,894 artists with both birth and death information were identified; institution/company records (305+ records detected via keywords such as “Company,” “Studio,” “Atelier”) and unrealistic lifespans (≤ 15 or ≥ 110 years) were filtered out, and then the top 1% of extremely prolific outliers (such as Mies van der Rohe, with 13,090 works) were removed, producing a model dataset of 5,656 artists.

Different threshold years (5, 10, 15, 20 years) and transformations (linear, log) were compared. As expected, model accuracy increased as the threshold year increased (R^2 : 5 years=0.313 $\rightarrow$ 20 years=0.798, linear), but a short threshold is needed for the model to be applicable to artists who died young (e.g. Basquiat, with an ~8-year career). The log transformation produced a clear improvement at every threshold (linear R^2 =0.705 at the 8-year threshold, versus only ~0.53 without the log transform); the final model was built with an 8-year threshold and a $\log(1+x)$ transformation:

$$\log(\text{total_works} + 1) = 1.057 \times \log(\text{first_8yr_works} + 1) + 0.182 \quad (R^2 = 0.705, n = 5,656)$$

The model was validated on two well-known artists who died young: for Keith Haring, the actual count of 49 works was predicted at 48.8 (nearly exact); for Jean-Michel Basquiat, the actual count of 12 works was predicted at 17 (+5 “lost potential”) — an expected deviation, since Basquiat's career was cut short right at the model's threshold (8 years).

The model was applied to 187 artists who died before age 40 and had a career of at least 8 years. The five artists with the highest “lost potential”:

Artist	Estimated lost potential (works)
Gordon Matta-Clark	+42.3
Rudolf Schwarzkogler	+34.3
Alberto Greco	+30.1
Robert Smithson	+24.3
Alfred Jarry	+20.8

Table 5. Artists with the highest estimated lost potential.

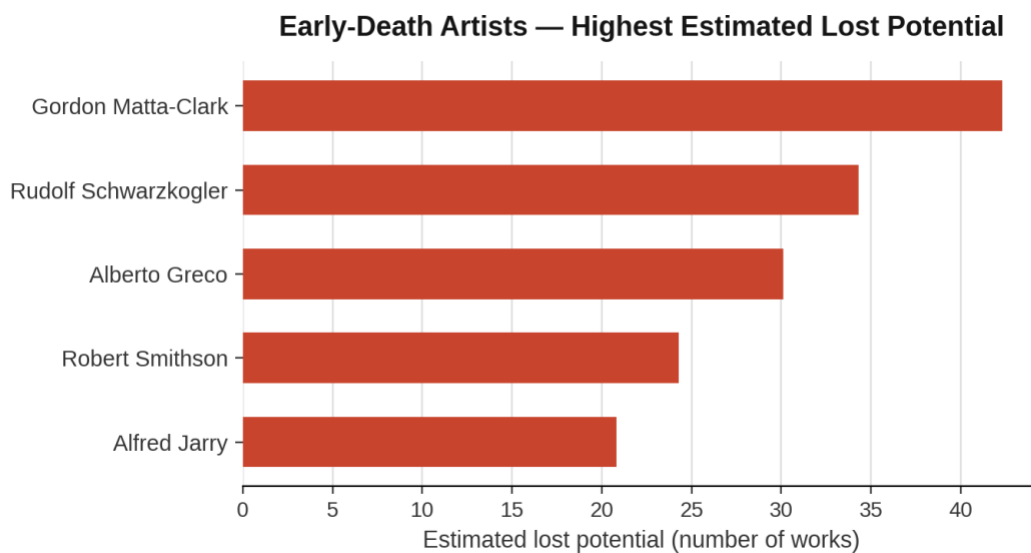


Figure 7. The five artists with the highest estimated lost potential.

4. Discussion

4.1 Robustness Check: The 1964 and 1968 Anomaly

A pairplot used to quickly scan relationships among numeric variables revealed that the distribution of acquisition year (acquired_year) was unexpectedly bimodal. Closer inspection showed this was not an organic trend but two individual mega-acquisition events: 19,010 works were acquired in a single year in 1968, and 12,987 in 1964 (roughly 10 times a normal year). Examining the content profile of these two years shows a

clear deviation from MoMA's overall average (83.0% male, 85.3% Western): in 1968, 99.3% of the works were by male artists and 98.8% by Western artists; in 1964, 98.0% were by male artists and 97.2% by Western artists.

This finding shows that individual “bulk acquisition” events can significantly affect aggregate statistics in institutional collection data; anomaly detection and robustness checks (comparing results with and without these events) are therefore an important methodological step in time-series analysis. As reported in the H1 section, the gender lag gap was largely preserved even after removing these two years, confirming that the finding does not depend on a single unusual event.

4.2 Methodological Lessons

Controlling for confounding variables

Raw findings can be misleading. H1's initially “in favor of women” finding could have led to a false conclusion (“MoMA positively discriminates in favor of women”) without a period-based comparison that holds production year constant. Applying the same control to H2, on the other hand, left the raw finding unchanged, confirming that the dynamics underlying the two hypotheses are genuinely different.

Getting the granularity right

Row count is not equal to unique artwork count. Averages and medians can be silently distorted if multi-artist and multi-movement relationships are not deduplicated at the `object_id` level. Before this check was applied in the H3 analysis, it was found that the same artwork repeated 49,546 times within the same movement group, and that the row count dropped from 128,013 to 72,360 after deduplication.

Testing model assumptions

The initial linear artwork-count-parity projection produced extreme and unrealistic results (such as the year 2350, or 1.8 million works). This stemmed both from the 2020 period being partial/incomplete in the dataset and from the 1960 outlier year distorting the mean. Using the median, removing outliers, and running a year-by-year simulation produced far more reliable estimates (2097, 2089). Similarly, in the early-death-artist model, using a log transformation instead of raw linear regression markedly improved model accuracy (R^2 from 0.31 to 0.70).

4.3 Limitations

- The H3 (movement) analysis covers only 43.4% of the collection (artists with a Wikidata record); this subset is not random and is likely biased toward better-known artists.
- Movement information belongs to the artist, not the artwork (Wikidata P135); the statement “this work is surrealist” actually means “this work's artist is associated with surrealism.”
- Lag differences measure MoMA's acquisition behavior; this alone cannot be claimed to reflect institutional “intent” — factors outside the data, such as market conditions, donation timing, and the number of surviving works, may also have played a role.
- All numerical projections (H1, H2, H4, early-death-artist) rest on the assumption that current trends continue unchanged; unforeseeable factors such as institutional policy changes could invalidate these estimates.

- The “institution detection” in the early-death-artist model was done via a keyword method and may not be 100% comprehensive.

5. Conclusion and Next Steps

This study systematically examined MoMA's collection acquisition behavior along four dimensions (gender, geography, art movement, temporal consistency) and put every finding through statistical validation. Key results:

- MoMA's gender-based acquisition-lag gap has closed completely, statistically — but achieving parity in artwork counts will require roughly 80 more years at the current trend.
- Inequality based on geographic origin narrowed over the periods but never lost statistical significance in any period — a more persistent pattern.
- MoMA's movement preference has reversed over time: it focused on its own contemporaries during its founding period, and is now compensating for movements it had neglected in the past.
- The institution's acquisition behavior is not random; it shows strong temporal consistency ($r=0.855$).
- Productivity in the first eight years of a career is a reasonable predictor of total output ($R^2=0.705$); this model gives concrete form to the potential gap in the MoMA collection for artists who died young.

The next step is to turn these findings into an interactive dashboard in Looker Studio / Power BI. All summary tables have already been prepared for this purpose in BigQuery's `moma_dashboard` dataset.

References

- MoMA Collection Dataset — github.com/MuseumofModernArt/collection (CC0 license)
- Wikidata — query.wikidata.org (SPARQL, CC0 license)